# Seth Strickland

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## Profile

Technical program lead and computational researcher who designs and ships NLP and computer-vision tooling and coordinates cross-functional research teams. Builds ML-backed systems end to end — data pipelines, transcription models, and LLM prompting and evaluation frameworks — and wins competitive funding through technical proposal writing, including an approved SBIR pre-application.

## Core Competencies

NLP & LLM Prompt Engineering • Transformers (BERT-like) & Computer Vision (ResNet) • OCR / Document AI (Kraken, Tesseract) • Data Pipelines & Tooling • Stylometry & Text Analytics • Product Design & Delivery • Cross-Functional Program Management • Technical & Grant Writing • Python, R, Julia, JavaScript • Git, Linux, LaTeX

## Experience

**Technical Liaison & Developer** **2023 – Present**
*Glyph Machina, Remote*

- Bridge domain experts and engineering to specify, build, and maintain software for a manuscript-transcription product, including a visual XML editor and automated Latin-abbreviation expansion.
- Authored an approved SBIR pre-application and internal technical documentation, opening a path to non-dilutive R&D funding.
- Own the end-to-end transcription workflow, from image line-segmentation through LLM transcription to validated, structured output.
- Trained a Kraken HTR model (CNN → LSTM) on ∼351,000 annotated manuscript lines to a 3% character-error rate on held-out validation, feeding an LLM correction pass; applied it to a new corpus of ∼40 medieval manuscripts (∼4,500 pages to date).

**Researcher** **2026 – Present**
*Proofs & Reasons, Carnegie Mellon University, Pittsburgh, PA*

- Lead an NLP and stylometry initiative modeling the long-run development of mathematical reasoning across a large historical corpus (proofsandreasons.io).
- Built an R/`stylo` classification pipeline that separates reasoning style from subject content, with leave-one-out cross-validation and distance-based reporting.
- Won a competitive NSF ACCESS allocation (1,454 GPU-hours + 1,399 GB on PSC Bridges-2) to train the lineation, transcription, and stylometry models on multi-century manuscript corpora.

**Creator & Developer** **2025 – Present**
*Research Party, Pittsburgh, PA*

- Designed and launched a bibliographic-analysis platform that maps social networks and bias across citation corpora and supports exploratory Q&A over a document set, now used by 24,330+ users across many labs/departments.
- Integrated an OCR-based citation-ingest pipeline for automated corpus construction.

**Founder & Coordinator** **2025 – 2026**
*Computational Humanities Initiatives, CMU, Pittsburgh, PA*

- Founded and ran cross-departmental research groups — one of which became a required graduate seminar — coordinating programming with the Posner Center and CMU Libraries.

- Managed events, logistics, and stakeholder coordination across departments and institutions.

**Lecturer, Writing & Communication** **2023 – Present**
*Carnegie Mellon University, Pittsburgh, PA*

- Designed a course uniting the philosophy of computer science and poetics, and built curricula on the critical, effective use of AI tools.

**Graduate Advisor (Special Faculty)** **Sep – Oct 2026**
*Carnegie Mellon University, Department of English, Pittsburgh, PA*

- Additional temporary part-time appointment (∼10 hrs/week): graduate recruitment and Dietrich application review; AI-assisted recruitment FAQs; PhD certificate launch coordination; handbook and RPT support; graduate data review.

## Selected Technical Projects

- **Research Party** — bibliographic-analysis platform I designed and launched — **see it live** at research-party.com; maps networks and bias across citation corpora with OCR-based ingest.
- **historical-ocr** — orchestrator turning historical sources into computation-ready text (OCR, IIIF/PDF ingest, YAML → TEI).
- **transcription-shell + transcription-protocol** — LLM transcription pipeline (125 commits) with a strict, provider-agnostic prompting protocol; benchmarks Anthropic, Gemini, and local Ollama models against ground truth.
- **manuscript-fingerprint** — computer-vision type attribution via Kraken line segmentation and a ResNet glyph encoder.
- **reasoning-stylometry** — R/`stylo` classifier separating mathematical reasoning style from content, externally validated against 31 human-declared labels.
- **strigil** — archival web scraper for high-quality text and images from digital-collection sources.

## Selected Research

- Co-author, "Democratizing the Medieval English Legal Tradition," *arXiv*, cs.CV (2026).
- "Making the Law Legible: A Self-Correcting BERT-Like Transformer as Manuscript Transcription Aid," Conference of the Association for the Study of Law, Culture, and the Humanities (2025).

## Education

**PhD, Medieval Literature** — Cornell University 2022

**MA, English** — Saint Louis University 2016

**BA, Classics & English Literature** — Hillsdale College 2012